MILITARY SPECIFICATION

BOOSTER, AMMUNITION, ELECTRIC, CALIBER .50

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers all electric ammunition boosters for the caliber .50, M3 machinegun.

1.2 **Classification.** Electric ammunition boosters for the caliber .50, M3 machinegun shall be one of the following types and classes:

Type I — Equipped with D. C. electric motor.

Class 1 — The booster will drive counterclockwise when viewed from the wire end and the projectile and sprockets will be next to the wire end.

Class 2 — The booster will drive counterclockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

Class 3 — The booster will drive clockwise when viewed from the wire end and the projectile end sprockets will be next to the wire end.

Class 4 — The sprocket will drive clockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

Type II — Equipped with A. C. electric motor.

Class 1 — The booster will drive counterclockwise when viewed from the wire end and the projectile end sprockets will be next to the wire end.

Class 2 — The booster will drive counterclockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

Class 3 — The booster will drive clockwise when viewed from the wire end and the projectile end sprockets will be next to the wire end.

Class 4 — The sprocket will drive clockwise when viewed from the wire end and the primer end sprockets will be next to the wire end.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

FSC 1005

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MIL–M–7969 — Motors, Alternating Current, 400 Cycles, 115/200 Volt System, Aircraft, General Specification for.

MIL–M–8609 — Motors, Direct Current, 218 Volt System, Aircraft Class A and B, General Specification for.

MIL–P–116 — Preservation, Methods of.

MIL–P–7936 — Parts and Equipment, Aeronautical, Preparation for Delivery.

STANDARDS

MILITARY

MIL–STD–129 — Marking of Shipments.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction Sample. As soon as practicable after award of contract or order and prior to submission of any complete unit for final acceptance, the contractor shall furnish two complete boosters of the type and class ordered, for inspection and test to determine conformance with this specification. The preproduction sample shall be the same in all details of design as the booster which the contractor proposes to furnish in fulfillment of the contract. Approval of the preproduction sample by the contracting officer shall not relieve the contractor of his obligation to supply boosters conforming to this specification. No change or deviation from the preproduction sample will be accepted without prior written approval from the contracting officer. Following approval by the Government Inspector, the preproduction sample may be delivered as one of the units under the contract.

3.2 Design. The ammunition booster assembly shall be designed for installation in ammunition boxes or elsewhere in the fire control system to aid the conveying of ammunition from the ammunition box to the feedway of the gun.

3.3 Reliability. The operational life of the unit shall be equivalent to a minimum of 84,000 rounds of ammunition passing over the booster when operated on a duty cycle per paragraph 4.2 of this specification.

3.4 Construction. The ammunition booster shall be a substantial compact unit so constructed that no parts will work loose in service. It shall withstand the strains, jars, vibrations and other conditions incident to shipping, storage, and service. The motor wires will come out from the end opposite the anti-roll-back release.

3.5 Maintainability. The ammunition booster shall be constructed to allow replacement of component parts with ease during overhaul or servicing.

3.6 Performance. The ammunition booster will meet the performance requirements as called out in the qualification test, Section 4, under the operating conditions therein stated.

3.7 Component parts. The ammunition booster shall consist of the following: an electric motor, two sprocket wheels, freewheeling clutch and anti-roll-back device assembled together within fixed end plates.

3.7.1 *Motor.*

Type I — The electric motor required shall be in accordance with the applicable requirements of MIL–M–8609 for intermittent duty motors.

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Type II — The booster shall be driven by a 200 volt, 3 phase, 400 cycle motor in accordance with applicable requirements of Specification MIL–M–7969 except as modified herein.

3.7.2 *Electrical properties.*

Type I — The booster at room temperature will not exceed a maximum of 16 amperes at 26 volts d.c. across the motor terminals when pulling 1400 rounds of ammunition per minute under a load of 50 pounds.

Type II — The booster at room temperature will not exceed a maximum of 2 amperes at 200 volts, 400 cycles, three phase alternating current across the motor terminals when pulling 1400 rounds of ammunition per minute under a load of 50 pounds.

3.7.3 *Sprockets.* The booster sprocket wheels will be arranged and placed as shown in Figure 1.

3.7.4 *Freewheeling Clutch.* The ammunition booster shall be constructed to allow the "housing assembly" to free wheel in the direction of drive with no indication of binding or seizing due to deformed bearings, soft bearing surfaces (raceways), mismatched parts or faulty workmanship.

3.7.5 *Anti-Roll-Back device.* The assembly will contain an anti-roll-back device to prevent booster rotation in a direction opposite to feeding rotation. This device will be spring loaded when normally connected and will be mechanically disconnectable for reverse rotation.

3.8 **Dimensions.** Dimensions for the ammunition booster shall be as shown in Figure 1.

3.9 **Weight.** Weight of the ammuntion booster shall not exceed five pounds.

3.10 **Identification of product.** Equipment, assemblies and parts shall be marked for identification in accordance with specification MIL–STD–130 using the following information:

Booster, Ammunition, Electric

Caliber .50, M3 Machinegun

Type Voltage System _____

Specification No.

Mfr's Part No.

Mfr's Serial No.

Order No.

Mfr's Name or Trademark

U.S. Property

3.11 **Workmanship.** All details of workmanship will be sufficiently high grade to ensure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

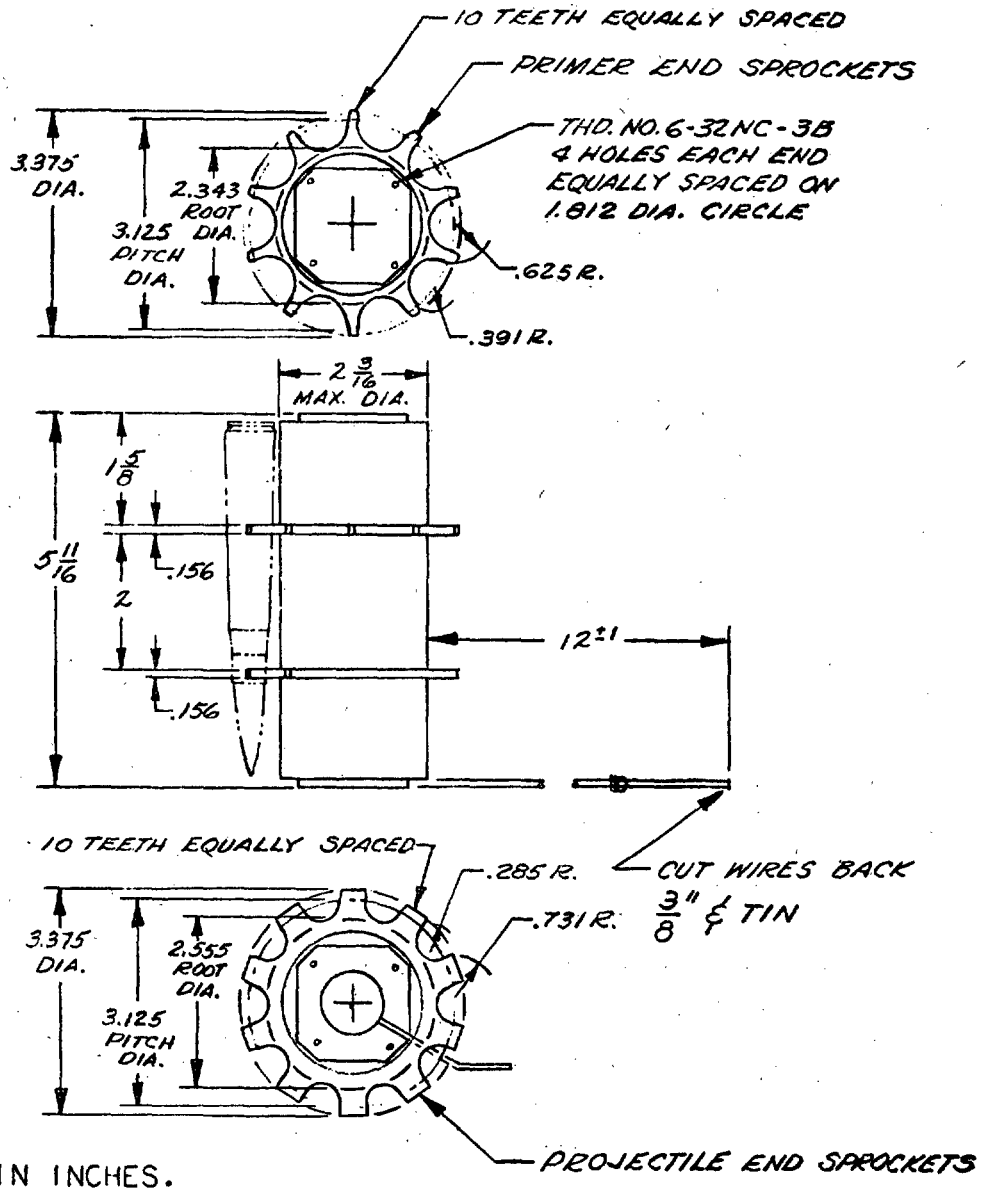
4.1 **Responsibility for Inspection.** Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to precribed requirements.

4.2 **Duty cycle.** The duty cycle shall be four seconds on, four seconds off, for four minutes of each hour.

4.3 **Classification tests.** The inspection and testing of booster shall be classified as follows:

4.3.1 *Preproduction tests.* Preproduction tests are those tests made on a sample representative of the production booster, to determine that the production equipment meets the requirement of this specification.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCE ON:
-ANGLES-
-DECIMALS-±.005
-FRACTIONS-±1/64

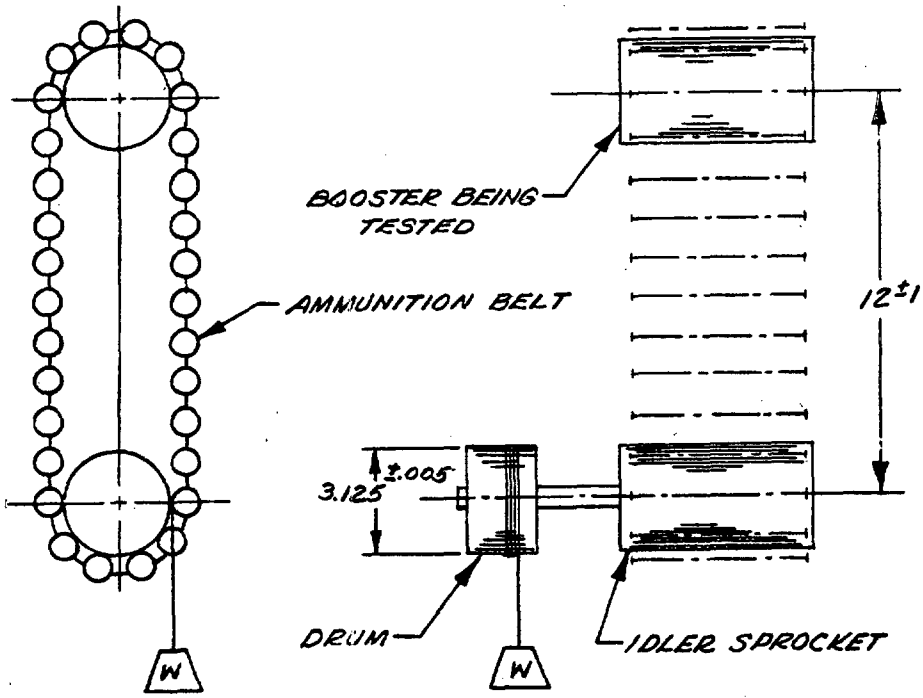
BOOSTER-AMMUNITION
ELECTRIC, 26V D.C. ; 200V A.C.
CALIBER .50

FIGURE 1

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THE WEIGHT OF THE IDLER SPROCKET DRUM AND CABLE SHALL BE AS LIGHT AS POSSIBLE TO REDUCE THE INERTIA LOAD

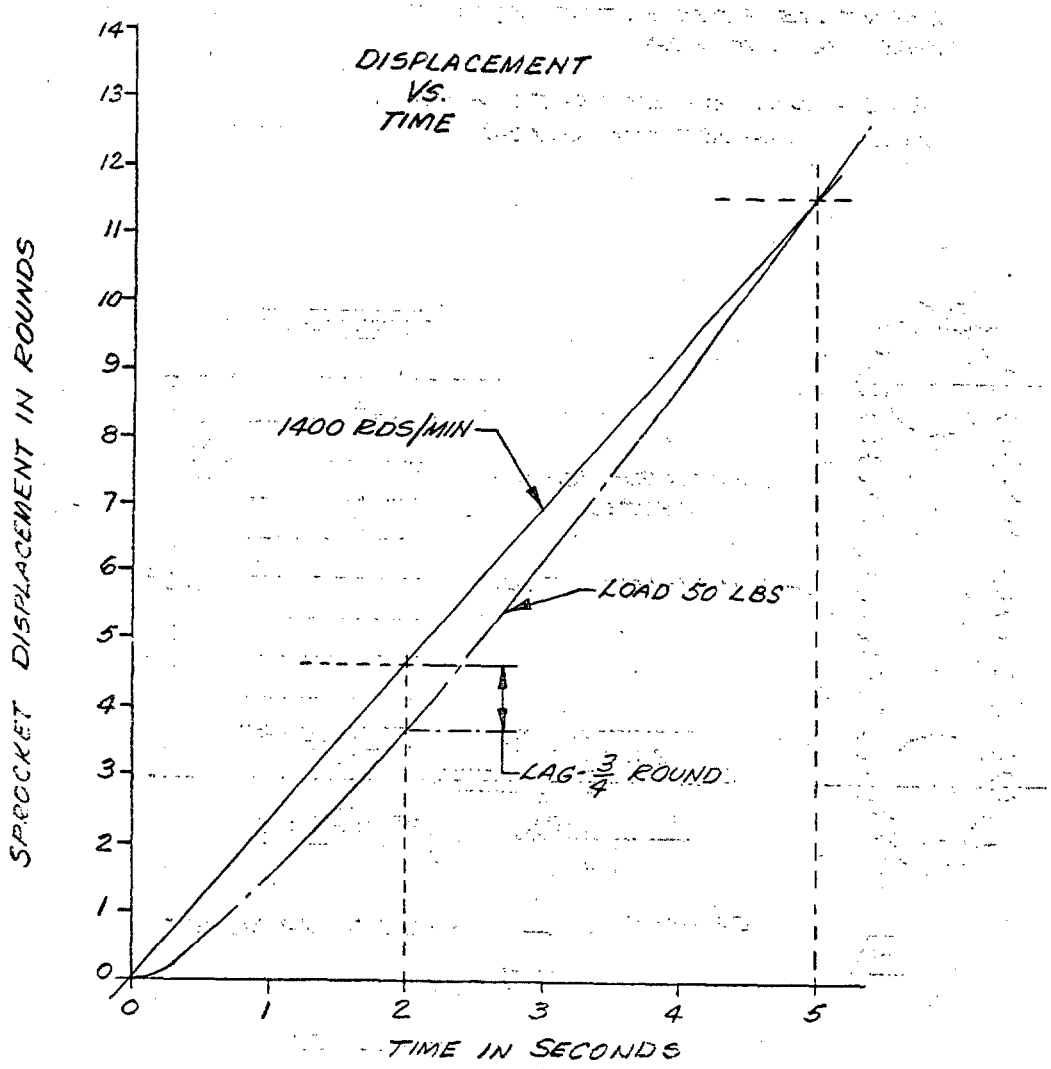
PITCH DIA. OF BOOSTER AND IDLER SPROCKET 3.125



BOOSTER TEST RIG

FIGURE 2

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GRAPH OF BOOSTER ACCELERATION WITH 50# LOAD

FIGURE 3

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4.3.2 *Inspection test.* Inspection tests are those tests accomplished on boosters submitted for acceptance under contract.

4.4 Preproduction tests.

4.4.1 *Preproduction test sample.* Two boosters of the type specified in the contract will be tested for design approval by the procuring agency, or when so specified in the contract, at the contractor's plant under the supervision of the procuring agency.

4.4.2 The preproduction tests shall consist of the following tests in the order they are given and those specified under inspection test method, subject to the test conditions specified herein. All tests are to be conducted.

4.4.2.1 *Cycle rate at room temperature.* The steady state of cycle rate with a 50 pound load shall be a minimum of 1400 rounds per minute when tested in accordance with a setup similar to the one shown in Figure 2.

4.4.2.2 *Cycle rate at room temperature.* After a 72 hour soak at –65° F and at an ambient of –65° F the booster cycle rate at 50 pound load shall be a minimum of 1200 rounds per minute throughout four duty cycles. After the completion of the –65° F tests, the booster shall be subjected to –100° F for at least one-half hour and at an ambient of –100° F the cycle rate shall be at least 1200 rpm with 50 pound load for one duty cycle.

4.4.2.3 *Stall characteristics.*

Type I — The sample shall be tested to determine that it will withstand 25 stalls of four seconds' duration spaced five minutes apart with no loss in efficiency when cooled at room temperature with power source of 28 volts d. c. across the motor terminals. The stalls may be accomplished by any applicable means.

Type II — The sample shall be tested to determine that it will withstand 25 stalls of four seconds' duration, spaced five minutes apart with no loss in efficiency when cooled at room temperature with power source of 200 volts, 400 cycles, alternating current across the motor terminals. The stalls may be accomplished by any applicable means.

4.4.2.4 *Load versus acceleration.* The sample shall be tested to determine its acceleration under rated load. The booster acceleration under a 50-pound load shall be such that in .20 second a cycle rate of 1400 rounds per minute shall be attained, at which time the booster shall not trail the gun fire rate (1400 rounds per minute) by more than 0.75 rounds. Further, it shall have caught up with the gun displacement within 0.50 seconds. Graphically the requirement shall be as shown in Figure 3.

4.4.2.5 *Anti-Roll-Back test.* The anti-roll-back device shall be able to withstand a 120 pound load at the pitch diameter. The maximum back-lash with a load of 45 pounds on the sprocket pitch diameter shall not exceed three degrees between the sprocket and the booster mounting frame. With the anti-roll-back in the release position, the booster sprocket shall roll back with a force not to exceed 6 pounds applied at the pitch diameter.

4.4.2.6 *Environmental.* These tests shall consist of those specified below of Specification MIL–E–5272.

Tests	Procedure
Humidity	I
Altitude	I
Salt Spray	III
Vibration	I
Fungus Resistance	I
Sand and Dust	I
Explosion Proof	I

4.4.2.7 *Life test.* The second sample shall be tested to determine capability of performing intermittent duty of four seconds "on", four seconds "off" for a period of four min-

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utes of each hour under a load of 50 pounds and at a rate of 1400 rounds per minute through a minimum of 3600 seconds operation, equivalent to 84,000 rounds of ammunition passing over the booster.

4.4.2.8 *Freewheeling clutch test.* The sample used in life test shall freewheel in the direction of drive when a force not to exceed 1 pound is applied to the pitch diameter.

4.4.2.9 *Inspection of preparation for delivery requirements.* Packing, packaging, and preservation shall be in accordance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** Unless otherwise specified, preservation and packaging shall be Level A and shall be accomplished in accordance with Method 11B of Specification MIL–P–116. Unit package quantity shall be one each.

5.2 **Packing.** Packing shall be Level A or B, as specified (see 6.2), and shall be accomplished in accordance with Specification MIL–P–7936, except that metal shipping containers shall not be used.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Use.** The type of ammunition booster covered by this specification is intended to be used to convey or assist in the flow of ammunition from the ammunition box to the caliber .50 machinegun.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, class, and number of preproduction samples, point of testing and delivery of preproduction test report.

(c) Submission of test programs for approval.

(d) Selection of applicable levels of packing.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Army—WC

Preparing activity:
Air Force—WRA

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